

WellComp 2023: Sixth International Workshop on Computing for Well-Being

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ABSTRACT

With the advancements in ubiquitous computing, ubicomp technology has deeply spread into our daily lives, including office work, home and house-keeping, health management, transportation, or even urban living environments. Furthermore, beyond the initial metrics commonly applied in computing, such as “efficiency” and “productivity”, the benefits that people (users) get from well-being-aware ubiquitous technology have been greatly emphasized in the recent years. Through the sixth “WellComp” (Computing for Well-being) workshop, we will discuss and debate the contribution of ubiquitous computing towards users’ well-being covering physical, mental, and social wellness (and the combinations thereof), from the viewpoints of various different layers of computing. Organized by a diverse international team of ubicomp researchers, WellComp 2023 will bring together researchers and practitioners from both academia and industry to explore versatile topics related to well-being and ubiquitous computing.

CCS CONCEPTS

• **Human computer interaction (HCI); • Ubiquitous computing;**

KEYWORDS

Well-being; Physical Wellness, Mental Health, Social Good

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1 BACKGROUND

In the advancing ubiquitous computing age, computing technology has already spread into many aspects of our daily lives, such as office work, home and house-keeping, health management, transportation, or even cities. We have been experiencing much of the influence of those technologies are both contributing to a better quality of life (QoL) of our individual and organizational lives, and causing new types of stress and pain at the same time. In the U.S., life satisfaction has not increased significantly for the last 30 years despite the gross wealth tripling, versatile computing technologies being introduced into their lives [8].

The term “well-being” has recently gained attention as a term that covers our general happiness and even more concrete good conditions in our lives, such as physical, psychological, and social wellness. Even economists and governments are starting to focus on well-being and “Gross National Happiness” as a new metric for measuring the statuses of the nations in addition to the traditional metrics such as GDP and unemployment rate. An increasing number of researchers, engineers, and people are paying attention to how their work can contribute to better quality of lives, social good, and well-being. This is quite in contrast to the criteria highlighted in the early age of computing, where the focus was on productivity and efficiency. Several trends in the technology field [1–4] are already being observed. Also, the “Positive Computing” community has initiated activities in the human computer interaction (HCI) domain [7], being inspired from positive psychology [9] that focuses on people’s psychology under healthy conditions (not in the ill-ed condition). Meanwhile, also in the UbiComp community, two workshops [5, 6] have took place, mainly focusing on detecting people’s mental illness by using smart and wearable devices.

In spite of such recent activities in academia and the society, unified academic research activities on computing and well-being is anticipated within the ubicomp research community. Active research not only in the HCI domain but in various other ubicomp research areas (systems, mobile/wearable sensing, mobile computing, persuasive applications and services, behavior change, etc.) are needed towards drawing the big picture of “computing for



Figure 1: Photos from WellComp 2018 (Oral Presentation Session (upper) and Poster Session (lower)).

well-being” from different viewpoints and layers of computing. Researchers have been already actively working on human activity recognition (mainly focusing on “physical” activities), psychological state recognition, and even societal activity recognition in urban and natural environments. Additional viewpoint of “well-being” of people/community in their researches will be possible with new types of applications that comprehensively cover such different types of activity recognition. Ever since Mark Weiser [10] introduced the term of ubiquitous computing, the ubiquity of computing in our daily lives and the society has been certainly progressing. Now it is time for the community to more seriously envision the benefits that such computing technologies can bring.

2 WELLCOMP 2023 WORKSHOP

In this workshop, we will bring together people from industry and academia who are active in the areas of activity recognition, mental health, social good, context-awareness and ubiquitous computing. Following five very successful workshops (WellComp 2018, 2019, 2020, 2021 and 2022) with more than 50 attendees respectively



Figure 2: Photos from WellComp 2019 (Keynote presentation by Prof. Philip J. Corr (upper) and Poster Session (lower)).

(shown in Figure 1 to Figure 4), the main objective of WellComp 2023 is to share the latest research in various areas in computing, related to users’ physical, mental, and social well-being. Expected areas and topics include, but not limited to, systems research, user studies, networking, social computing, as shown in the list as follows:

- Measurement and representation of physical, mental, and social well-being with ubicomp technologies.
- Design and implementation of platforms for collecting, processing, and interpreting health and well-being data.
- Design and development of computational models predictive of one or several aspects of well-being.
- Unsupervised, semi-supervised, and supervised representation learning for well-being.
- Classification, regression, and clustering problems related to well-being aspects.
- Approaches addressing challenges in wearable sensor data (e.g., missing and noisy data, irregular sampling rates, few labels, out-of-distribution inputs, etc) used for well-being.

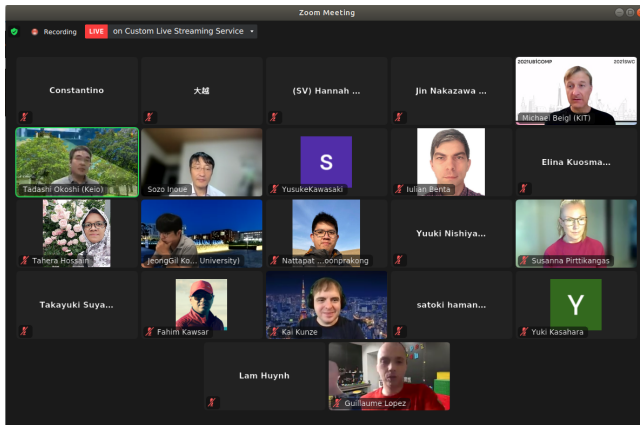


Figure 3: Screenshot from the virtual meeting at WellComp 2021 (Virtual Event).

- Development of explainable, robust, privacy-aware, and trustworthy pipelines for well-being monitoring.
- Multi-modal approaches integrating information from several data sources (e.g., physiological, behavioral, audio).
- Computing systems for promoting well-being-awareness.
- Innovative well-being applications and diverse target populations (e.g., children, patients, or elderly people).

Especially for the sixth workshop, this year’s special attention will be drawn to papers addressing the “challenges for physical, social and mental well-being monitoring using ubicomp technologies.” Relevance to such topics will be considered in the paper review and selection process. Furthermore, the workshop aims to identify future research challenges, research opportunities, and applications of our research outcomes to society.

Relevance and impact to UbiComp: The emergence of versatile types of computing devices (including IoT), services, and computing venues has been realizing the advanced “ubiquity” of computing. WellComp 2023’s focus on user well-being is related to all of the UbiComp research fields above and can open new applications to which ubicomp technology can be applied. Furthermore, the introduction of new applications and insights from the WellComp workshop also has significant potential to impact and contribute to UbiComp’s interest areas horizontally.

Long-term objectives: The workshop will lead to a deeper understanding of well-being and its relationship with ubiquitous computing. The sixth workshop will position the theme within the UbiComp research community and the definition of “computing for well-being” will continuously be refined. Under such efforts, we plan to build an active and long-lasting community around the workshop’s theme. For example, we want to use the workshop’s drive to prepare a special issue of a journal, e.g., IEEE Pervasive Computing. We plan to make an open call for this special issue, but will especially invite workshop participants to submit their work. Furthermore, we consolidate what we find from the workshop (including discussion) into an article that highlights the workshop’s main insights.



Figure 4: Photos from WellComp 2022 in Cambridge, UK.

The workshop’s website, along with social media accounts, will be the central venue for us to disseminate such insights from the workshop. Our website is available in <https://wellcomp-workshop.github.io/2023/> domain. We will share all workshop contributions (including papers and discussion results) on this website. Furthermore, social media pages (e.g., Facebook page) will be leveraged to reach potential authors, participants, and further collaborators.

3 ORGANIZERS

WellComp 2023 has a strong international organizing team of researchers in versatile ubicomp-related areas.

- **Shkurta Gashi** is a postdoctoral fellow at the ETH AI Center, where she is working on the intersection of ubiquitous computing, machine learning, and digital health. She obtained her Ph.D. degree in Informatics, from Università della Svizzera italiana (USI), Switzerland. In 2021, she has been a visiting researcher at the Nokia Bell Labs in Cambridge. She has previous experience organizing scientific meetings such as UbiComp/ISWC 2022 (web chair), UbiComp WellComp

2022 (co-organizer), and PerCom 2018 N2Women Event (organizer).

- **Dimitris Spathis** is a research scientist at Nokia Bell Labs, Cambridge (UK) and a visiting researcher at the University of Cambridge. His work enables AI to make the most out of real-world multimodal and sequential data through label-efficient and robust ML. He previously worked at Microsoft Research, Telefonica Research, Ocado, and Qustodio. His experience as an organizer of scientific meetings includes WellComp at UbiComp 2022 (general chair), CHIL 2023 (track chair), Federated Sensing tutorial at MobiCom 2021 (co-organizer), and ML4H co-located with NeurIPS 2022 (senior roundtable chair).
- **Ting Dang** is a senior research scientist at Nokia Bell Labs, Cambridge (UK), and a visiting researcher at the University of Cambridge. Her research is focused on human-centered audio sensing and machine learning for mobile health monitoring (e.g., emotion, depression, COVID-19). She has experience in the successful bidding for Interspeech 2026.
- **Alexander Hoelzemann** is a doctoral student at the chair of Ubiquitous Computing at the University of Siegen, Germany. His research focuses on human activity recognition with wearable sensors. This includes research towards data recordings and annotation processes as well as machine and deep learning techniques. He is currently helping to organize UbiComp/ISWC 2023 as one of the web chairs and has been a student volunteer chair at UbiComp/ISWC 2020.

The Steering Committee of the WellComp Workshops is:

- **Tadashi Okoshi**, Associate Professor, Keio University, Japan
- **Jin Nakazawa**, Associate Professor, Keio University, Japan
- **JeongGil Ko**, Assistant Professor, Ajou University, South Korea
- **Fahim Kawsar**, Department Lead, Nokia Bell Labs Cambridge, UK
- **Susanna Pirttikangas**, Research Director, University of Oulu, Finland

4 WORKSHOP PAPERS

Section 2 shows the possible topics of the papers we will ask the workshop authors to contribute. We will accept two types of papers *long* (6 pages) and *short* papers (4 pages) excluding the references in the double-column format. Both papers should use the ACM SIGCHI Master Article template suggested in the UbiComp/ISWC 2023 submission guidelines¹. The accepted papers will **be included in the ACM Digital Library**. We will distribute our call for papers to various mailing lists of ubicomp-related researcher communities, including ubiquitous, pervasive and mobile computing, HCI, systems research, IoT, distributed systems, or mobile health.

At least two reviewers from our organizers and PC members will peer-review each submission. Our acceptance criteria will be a mixture of relevance, novelty, provocativeness, and research quality. **We expect to have a selection of 10-15 accepted papers.** Given the theme of this workshop, we are confident to attract a large number of paper submissions which enables us to construct a high-quality workshop. We will request at least one author of each

Table 1: The expected workshop program: a mixture of inspiring keynotes, and interactive paper presentations.

Time	Program
09:00–09:15	Introduction
09:15–10:15	Keynote Talk 1
10:15–10:30	Coffee Break
10:30–11:30	Paper Presentations Session
11:30–12:30	Keynote Talk 2
12:30–12:45	Closing Remarks & Future Planning
12:45–13:30	Networking & Lunch Break

accepted paper to register for the conference and the workshop itself. All papers will need to be anonymized.

5 WORKSHOP STRUCTURE

Following five successful workshops in 2018, 2019, 2020, 2021 and 2022 we plan WellComp'23 as a **half-day workshop** whose program is shown in Table 1. We're planning to have two keynote lectures by distinguished researchers focusing on "well-being", inspired by the successful and interactive keynotes from WellComp 2022. The first keynote will be followed by the paper presentation session and intensive Q&A sessions. During the paper presentation session, each accepted paper will be presented by one of the authors. The closing session will summarize the key findings from the workshop, and discuss possible future collaborations and activities of the community. We will then allow the workshop participants to have a networking and interacting session at the end of the workshop. Participation in the workshop will not be limited.

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¹Guidelines: <https://www.ubicomp.org/ubicomp-iswc-2023/authors/formatting/>